EXP NO:-6

PYTHON IMPLEMENTATION OF HAMMING CODE CONCEPT

Error Correction at Data Link Layer (AIM):-

→ Hamming Code is a set of error-correction codes that can be used to detect and correct the errors when the data is transmitted from the sender to the reciever. To implement error detection & correction using Hamming code technique Data Link Layer.

SENDER PROGRAM:-

- → Takes text input
- → Converts text to binary
- → Divides binary data into 4-bit groups.
- → Add 3 redundant parity bits using hamming code
- → Saves the encoded data in chanel txt.

CODE:-

```
def hamming-encode (data):
    encoded = ""
for i in range (0, len(data),4):
    d = data [i:i+4]
    if len(d) <4:
        d = d.ljust (4,'0')
    d1, d2, d3, d4 = map(int,d)
    P1 = d1 ^ d2 ^ d4
    P2 = d1 ^ d3 ^ d4
    P4 = d2 ^ d3 ^ d4
code = str(P1) + str(P2) + str(d1) + str(P4) + str(d2) + str(d3)
```

```
return
text = input ("Enter text: ")
binary = ''.join (format (ord (c), '08b') for c in text)
print ("Binary data":, binary)
encoded_data = hamming_encode (binary)
with. open ("channel.text", "w") as file:
  file.write (encoded_data)
print ("Hamming encoded data: ", encoded_data)
print (" Data saved to channel.txt")
```

RECIEVER PROGRAM:-

→ Reads data from channel.txt

→ Checks every 7 bit hammering block

→ Find the error position using syndrome

→ Corrects the errors & removes parity bits

→ Converts binary back to ASCII text

Code:-

```
def hamming_decode (data):
  corrected = ""
  error_found = False
  for i in range (0, len(data), 7):
    block = list(map (int, data [i : i+7]))
    if len (block) < 7:
      continue
    p1, p2, d1, p4, d2, d3, d4 = block
```

~~S4 = p4 ∧ d2 ∧ d3 ∧ d4~~

```
S4 = p4 ∧ d2 ∧ d3 ∧ d4
error_position = s1 + (s2 * 2) + (s4 * 4)
if error_position ! = 0
  error_found = True
  actual_position = i + error_position
  print ["Error detected at bit position:", actual_position
corrected += str(block[2])
corrected += str(block[4])
corrected += str(block[5])
corrected += str(block[6])
return corrected, error_found
with open("channel.txt", "r") as file:
  channel_data = file.read(). strip()
print("Data recieved from Channel:")
print(channel_data)
binary_data, error_found = hamming_decode(channel_data)
text = " "
for i in range(0, len(binary_data), 8):
    byte = binary_data [i: i+8]
    if len[byte) == 8:
        text += chr(int(byte, 2))
print("Incorrected binary data:")
print(binary_data)
if error_found:
    print("\nError was detected and corrected")
else:
    print("\n No error detected")
print("Recieved text: ", text)
```

OUTPUT:-

~~Print~~ Enter the text: HELLO

OUTPUT:-

```
Binary data: 01001 000010001 0101001 00010011
             0001001111

Hamming encoded data = 1001100 11100 000100110 00
                       10010100111000 11110 0100 111111

Data saved to channel.txt
```

RESULT:-

Thus the hamming code was successfully implemented to detect & correct errors in the transmitted data.